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Impact of occupant-centric thermal control on operational carbon performance for building sustainability

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Abstract

This study evaluates the effectiveness of occupant-centric control for enhancing carbon performance during building operation, extending beyond the conventional objective of maintaining thermal comfort. An occupant-centric control framework is proposed that integrates real-time metabolic rate (MET) and clothing insulation (CLO) into the HVAC control system through a feedback-based decision-making loop. The methodology follows a three-stage evaluation process: building energy simulations examining 110 combinations of MET (0.7–1.7 met) and CLO (0.1–1.0 clo); controlled test-bed experiments with eight participants across nine MET–CLO conditions; and an eight-day living lab validation with 11 occupants. All stages were evaluated against a conventional temperature-based control baseline, and carbon emissions were quantified from simulated and measured heating and cooling energy consumption using source-specific emission factors. Results show that dynamic occupant variables significantly influence system-level carbon emissions, with MET exerting a greater impact than CLO under heating conditions. In the living lab, the proposed framework reduced carbon emissions by approximately 63% on average relative to conventional control, while maintaining target comfort levels, and emission patterns observed in simulations were consistently reproduced in field experiments. These findings indicate that occupant-related variables function as critical control inputs shaping operational carbon performance, underscoring the potential of occupant-centric control as a scalable strategy for low-carbon building operation.

Keywords

Occupant-centric control, human-in-the-loop, operational carbon performance, building sustainability, thermal comfort

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Introduction

The building sector is a major contributor to global carbon emissions, accounting for approximately 38% of all emissions, according to the International Energy Agency.[1] Reducing carbon emissions from buildings is essential for mitigating greenhouse gases and achieving carbon neutrality. Notably, energy use and carbon emissions during the operational phase of buildings account for over 70% of the building sector's total emissions.[2] Therefore, improving energy efficiency and cutting carbon emissions during the operational phase of buildings are critical steps toward operational decarbonization and the transition to a smart and sustainable built environment.[3]

Numerous studies have explored carbon emissions related to building operations.[4–10] In real-world building use, various external factors affect carbon emissions, making it essential to develop strategies to manage them effectively.

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Consequently, many studies are being conducted to address these challenges through digital transformation and data-driven optimization of building systems.[4,11–13] Recently, efficient control technologies for building systems have advanced through the development of intelligent, AI-enabled control systems and proposals for predictive and adaptive control frameworks.[11–13]

In the operational phase of buildings, occupant-related information has increasingly been recognized as a critical factor in achieving efficient building performance. Unlike system-centric control approaches, incorporating dynamic occupant information can reduce unnecessary energy use and mitigate occupant discomfort, particularly in shared or multi-occupant spaces. As a primary control indicator among various occupant-related variables, thermal comfort critically influences both occupant well-being and building operational efficiency, necessitating an accurate assessment to ensure sustainability within smart, adaptive building systems. Sustainability performance has also been shown to vary considerably across building typologies due to differences in climate response, building characteristics, and occupant behaviour, highlighting the importance of context-specific and occupant-responsive approaches for improving building performance.[14,15] Since occupants' thermal demands directly dictate HVAC performance, intelligent control strategies must integrate precise comfort evaluation as a foundational element of operational decision-making.

To evaluate thermal comfort, this study adopted the predicted mean vote (PMV) model, which is widely utilized in international standards.[16,17] To calculate PMV, personal factors, including metabolic rate (MET) and clothing insulation (CLO), are required in addition to indoor environmental factors such as indoor temperature (T_{air}), humidity (RH), mean radiant temperature (T_{mrt}), and air velocity (V_{air}). These personal variables are dynamic, context-dependent inputs essential for occupant-centric control and operational decarbonization. MET represents the heat produced by the human body during physical activity and is measured in met units (1 met = 58.1 W/m2). CLO indicates the insulation value of the clothing worn, which varies with the garment type and combination, and is measured in clo units (1.0 clo = 0.155 m^2K/W).

Accurate measurement of occupant thermal comfort requires real-time estimates of MET and CLO, which can vary over time. Traditionally, no simple method has been developed for objectively measuring these variables without occupant input. In practice, these variables are often disregarded or assigned arbitrary values,[17–19] limiting the effectiveness of occupant-centric and data-driven building operations. Recent advancements in sensing and computational technologies have begun to address these limitations. As part of the broader digital transformation of the built environment, real-time thermal comfort estimation has been enabled by new thermal comfort models[20,21] and methods for extracting MET and CLO values from indoor images.[22–26] Notably, progress in artificial intelligence (AI) has enabled the development of computer vision-based deep learning models that can be trained on indoor images to estimate thermal comfort in real time.[26,27] For example, Choi et al.[26] developed a computer vision-based AI model that detects individuals, their body poses, and clothing to estimate MET and CLO from a single indoor image. This method operates without human intervention and enables the objective and continuous derivation of dynamic occupant variables.

Recent technological advancements have paved the way for human-centric, adaptive thermal comfort control by leveraging real-time occupant state data. A significant body of research has explored building control systems that utilize occupant-derived real-time PMV as a key regulatory parameter.[27–29] These investigations evaluate how PMV-based control influences thermal environments and verify the robustness of these models when applied to operational building systems. Collectively, these developments are driving the transition from conventional system-centric paradigms to adaptive, occupant-centric building management strategies.

Despite these advances, achieving both thermal comfort and carbon emissions reduction simultaneously remains a significant challenge in operational decarbonization.[30–32] For example, Wang et al.[4] proposed an HVAC control framework aimed at minimizing carbon emissions, and Yang et al.[11] focused on optimizing energy consumption and reducing carbon emissions through system-level integration and operational control strategies. However, in these studies, thermal comfort indicators were not integrated into the optimization frameworks; instead, they were used solely to evaluate the effectiveness of carbon reduction strategies. Similarly, Magnier and Haghighat[12] developed a multi-objective optimization method to balance building energy consumption and thermal comfort, but did not consider dynamic occupant thermal comfort.

Notably, energy consumption alone does not fully capture the environmental impact of building operations, as buildings typically rely on multiple energy sources, such as electricity and fossil fuels, that differ substantially in carbon intensity per unit of energy. Consequently, even a control strategy that reduces total energy consumption may not achieve a proportional reduction in carbon emissions if it alters the relative share of these energy sources. This distinction is particularly relevant for occupant-centric control, where dynamic adjustments in heating and cooling operation can shift the balance between electricity-based and fuel-based energy use. Carbon emissions therefore represent a more policy-relevant and technically appropriate metric than energy consumption alone for evaluating the decarbonization potential of occupant-centric control strategies.

In summary, optimal building control requires a multi-objective approach that accounts for the trade-offs among thermal comfort, energy consumption, and operational carbon emissions. The influence of dynamic occupant variables on thermal

comfort has been well established through prior research on occupant-centric control and model-predictive control.[26,27,33] However, few studies have quantitatively examined how diverse occupant states affect actual building operation in a systematic, staged manner. In particular, limited attention has been given to how sensitive operational carbon emissions are to MET and CLO individually when dynamic occupant variables are integrated into a control system in real time, and how this sensitivity differs between the cooling and heating seasons. Disaggregating the contribution of these key occupant variables to operational-phase carbon emissions at the level of individual variables is therefore also necessary, as it provides foundational data for the future design of multi-objective control strategies that jointly consider thermal comfort and carbon emissions.

Accordingly, this study conducts a sensitivity analysis of carbon emissions associated with changes in occupant MET and CLO, disaggregated by energy source, and extends prior occupant-centric control research by validating these findings through a three-stage framework encompassing simulation as well as field experiments (test-bed and living lab). The variable-specific carbon sensitivity data derived in this study can serve as empirical evidence for the future design of multi-objective control strategies that jointly optimize thermal comfort and carbon emissions.

Conceptually, this study frames occupant-centric control as a human-in-the-loop feedback system in which dynamic occupant states (MET and CLO) function not as passive comfort parameters but as active state variables that propagate through the control decision-making process to shape system-level operational and environmental outcomes. Specifically, changes in occupant thermal state alter the calculated PMV, which in turn adjusts the control setpoint, thereby modifying HVAC operational behaviour and, consequently, the magnitude and composition of energy consumption and associated carbon emissions. Accordingly, this study extends the conventional occupant-centric control framework by explicitly linking occupant dynamics to operational carbon performance through the building control process.

This study addresses the identified research gap by proposing and implementing a thermal comfort–based building control framework that integrates real-time dynamic human factors. Specifically, an occupant-centric thermal control framework was developed to incorporate dynamic occupant MET and CLO into an adaptive HVAC control loop and evaluate its impact on operational carbon emissions. To ensure methodological rigour, the study adopts a three-stage evaluation framework that bridges digital modelling and real-world validation. First, simulation-based sensitivity analyses were conducted to systematically examine multiple MET–CLO combinations and quantify carbon emission responses under varying occupancy conditions. Second, controlled test-bed experiments were performed to validate the operational behaviour of the proposed framework under defined environmental scenarios. Finally, a living lab study in an actual office environment collected real-time occupant data alongside corresponding operational carbon performance metrics to assess the feasibility and effectiveness of the proposed human-centered control strategy in real building operation.

Overall, this study provides quantitative evidence of how dynamic occupant information can influence system-level carbon performance across both simulated and real operational contexts. By embedding real-time occupant conditions into control decision-making, the proposed framework demonstrates that occupant-aware strategies can simultaneously enhance thermal comfort and reduce carbon emissions. These findings highlight the value of interactive, occupant-centric control strategies as a practical operational pathway toward smart and low-carbon building performance.

Methods

This study investigates the impacts of integrating dynamic human information into building control systems. The proposed methodology is organized into three main steps, as illustrated in Figure 1. In Step 1, an AI–based recognition approach is introduced to estimate key human variables, MET, and CLO. An occupant-centric HVAC control framework that incorporates these variables into feedback control of a HVAC system is also described. The detailed processing pipeline is presented in the *Occupant-centric Thermal Control Framework*.

In Step 2, a sensitivity analysis is conducted to investigate how human variations affect system operation and carbon emissions before on-site experimentation. The purpose of this stage is not to analyze absolute operational outcomes but to explore how dynamic occupant information drives changes in operational carbon performance. To this end, a set of simulation scenarios is constructed using an EnergyPlus-based building model, in which physical and environmental conditions are held constant while MET and CLO values are systematically varied within a limited range. Operational carbon emissions are quantitatively analyzed across these scenarios. The results provide a foundational design space for subsequent experimental design and interpretation.

In Step 3, on-site experiments are conducted to validate the effects of real-time, occupant-centric building control in operational environments. Here, validation does not imply replicating the physical scale or configuration of the EnergyPlus reference building used in Step 2; rather, it means examining whether the qualitative sensitivity patterns identified in the simulation are directionally reproduced under physically distinct real-world system conditions. First, a small-scale test-bed experiment is carried out under controlled conditions with a single occupant to examine the fundamental control response

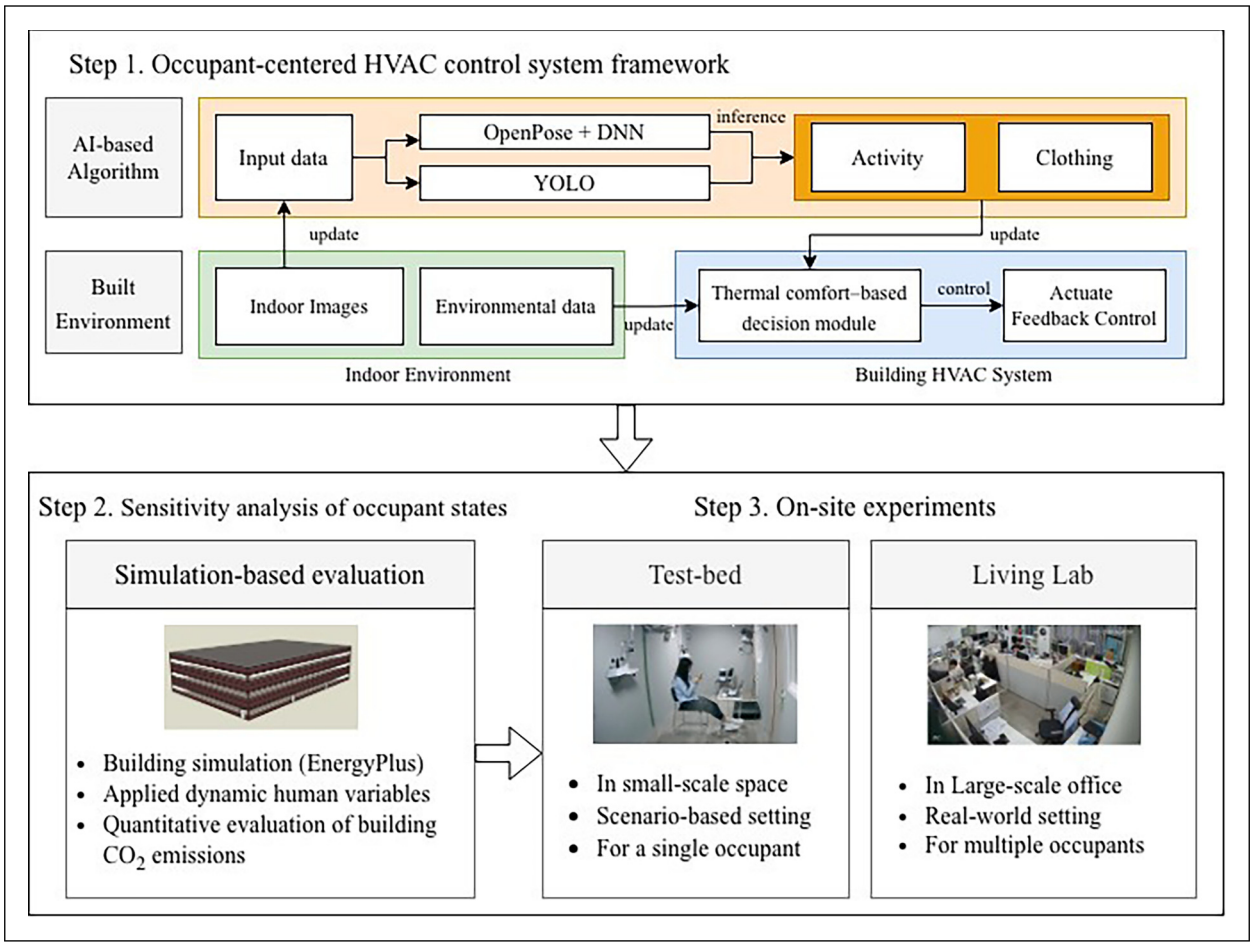


Figure 1. Research framework.

characteristics. Subsequently, empirical evaluations are conducted in a living lab environment with multiple occupants. Through this process, the study analyzes how human state–based control influences the actual operation of HVAC systems under realistic conditions. This stage serves as an empirical validation that bridges simulation-based digital analysis and real-world building operation. It demonstrates how integrating dynamic occupant information into system decision-making leads to observable changes in system behaviour and corresponding impacts on operational carbon emissions. Detailed descriptions of the implementation procedures, evaluation metrics, and experimental settings for each stage are provided in the following sections.

Occupant-centric thermal control framework

This section describes a system framework designed to enable real-time building control with occupants' states in operational environments. As illustrated in Step 1 of Figure 1, the proposed framework consists of two main components: an AI–based algorithm for recognizing occupant states and a physical built environment component that integrates the recognized human information into actual building systems.

The core algorithm employed in this study is a computer vision–based AI model previously developed by Choi et al.[26] In this study, this algorithm is integrated with a real building HVAC system and applied across various occupancy scenarios to evaluate the system-level performance of occupant-centric building control. The algorithm uses real-time indoor images collected from the operational environment as input data and outputs occupant-specific information, including MET and CLO.

The developed algorithm adopts a modular pipeline architecture. The overall process is divided into three main stages. In the first stage, human body features are extracted from indoor images to generate foundational information for interpreting occupant states. This stage is implemented using an OpenPose-based pose estimation method.[34] Specifically, occupants

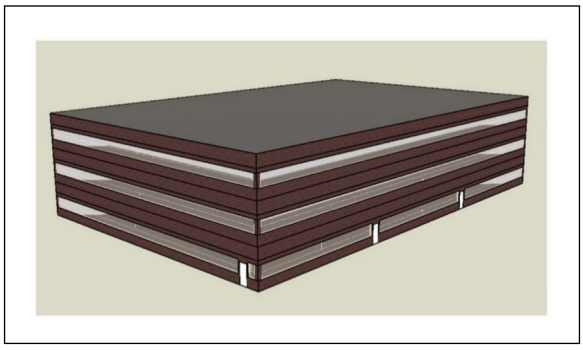


Figure 2. Model image.

are detected in the indoor environment, and joint keypoints for each individual are extracted to construct a quantitative representation for subsequent human-state analysis.

In the second stage, occupants' activity levels are inferred based on the extracted pose information. Keypoint-based feature representations serve as inputs to a deep neural network (DNN) classifier, which maps current physical posture and movement patterns to meaningful human MET variables. In the third stage, clothing information is recognized to estimate CLO. This process is performed using a YOLOv5-based object detection model that identifies clothing worn by occupants and provides additional human-state information relevant to thermal assessment.[35] The models employed in this framework were selected not to introduce algorithmic novelty, but to ensure robustness and real-time applicability in indoor environments. Detailed descriptions of the network architectures and training procedures are provided in prior work[26] and are therefore not repeated here.

The developed model can estimate five discrete MET levels and up to 34 CLO values from combinations of clothing ensembles. The system operates in real time, enabling continuous estimation of personal factors. Validation experiments with human participants demonstrated that using the most frequently observed values, accumulated over periods of more than 5 min, yields highly accurate MET and CLO estimates. Performance evaluation using the F1 score, the harmonic mean of precision and recall, showed strong performance, with F1 scores exceeding 0.93 for both MET and CLO predictions. Here, precision is defined as the proportion of correctly classified samples among all samples predicted to belong to a given activity level or garment category, whereas recall is defined as the proportion of correctly classified samples among all true samples of that category. The F1 score, calculated as the harmonic mean of precision and recall, provides a balanced measure of classification performance across the five MET activity levels and the garment categories used to estimate CLO. An F1 score approaching 1 indicates near-perfect classification performance.

The recognized occupant state information is subsequently used as input to the building HVAC system's decision-making process. Indoor environmental data are combined with occupants' estimated MET and CLO to represent the current thermal state using the PMV index. The calculated PMV serves as a reference for defining the discrepancy between occupants' current and target thermal states and functions as a key variable in HVAC control decision-making.

Simulation setup

The simulation-based analysis was designed to evaluate the sensitivity of HVAC system responses to variations in occupants' personal factors. It systematically changed the MET and CLO while keeping the physical building configuration and operational schedule constant to analyze the impact of occupants' factors. To this end, EnergyPlus was used to implement PMV-based control incorporating MET and CLO as inputs, based on a standardized medium office reference model compliant with ASHRAE Standard 90.1.[36,37] PMV-based control was implemented in EnergyPlus using the ZoneControl:Thermostat:ThermalComfort object together with the ThermostatSetpoint:ThermalComfort:Fanger:SingleHeatingOrCooling object. A target PMV setpoint of 0 was applied consistently across all simulation cases.

As shown in Figure 2, the building is a three-story office structure with a total floor area of 4982.19 m^2. The key modelling parameters are listed in Table 1. To simulate carbon emissions during cooling and heating periods, the weather file for Seoul, South Korea, was used. This location was selected to evaluate the effects of cooling and heating periods, as it is classified as Climate Zone 4A under ASHRAE Standard 169, which is characterized by significant seasonal variations.[38] Accordingly, the heating period was set for January, February, November, and December, and the cooling period was defined as June, July, August, and September, excluding transitional seasons.

Table 1. Modelling information.

Modelling	Information	Modelling	Information
Basic information		*Internal Load*	
Target model	Office Medium (ASHRAE 90.1)	Occupancy	18.58 m^2/people
Location	Seoul, South Korea	Activity Level	0.7–1.7 met (72–180 W/person)
Weather	4A (mixed humid climate)	Clothing Insulation	0.1–1.0 clo
Number of Floors	3	Office equipment	8.07 W/m^2
Ground floor	4982.19 m^2	Lighting	400 lux (6.89 W/m^2)
Wall U-value	0.704 W/m^2K	*HVAC system*	
Floor U-value	0.704 W/m^2K	Heating system	Gas heating and electric reheat
Roof U-value	0.358 W/m^2K	Cooling system	Direct expansion
WWR	33.0%	Fan system	VAV
Window U-value	3.24 W/m^2K	Cooling COP	3.4
Window SHGC	0.39	Burner Efficiency	0.81
		Ventilation	1.14 $L/s{\cdot}m^3$
		Cooling period	6/1-9/30
		Heating period	1/1-2/28, 11/1-12/31

Occupants are considered present from 8 AM to 7 PM, reflecting typical office usage patterns. The HVAC system features a packaged air conditioning unit as the central facility, supported by an Air Handling Unit (AHU) system and Variable Air Volume (VAV) boxes installed in each zone. The chiller used for the AHU's heating and cooling coils has a coefficient of performance (COP) of 3.4, and the boiler's efficiency is 0.81. The system's heating coil uses natural gas as the energy source, whereas the cooling and reheat coils of the VAV boxes are powered by electricity. Humidity was not actively controlled in the simulation, consistent with the default configuration of the ASHRAE 90.1 medium office reference model, which does not include a dedicated humidification or dehumidification system. Instead, indoor relative humidity varied dynamically as a result of the psychrometric process of the direct expansion cooling coil. The simulated relative humidity at each timestep was used as an input for the PMV calculation.

The analysis was conducted from multiple combinations of MET and CLO conditions by modifying the occupant variables. The ranges of MET and CLO were determined based on typical indoor office conditions and feasible control operations. MET values were varied from 0.7 to 1.7 met in increments of 0.1 met (11 levels), and CLO values were varied from 0.1 to 1.0 clo in increments of 0.1 clo (10 levels), resulting in 110 combinations. CLO ensemble examples used to define the insulation levels are provided in Appendix 1 (Table 4). These simulations provide a controlled baseline for interpreting system responses to variations in occupant state before on-site evaluation.

On-site experimental setup

This section describes the experimental settings used to evaluate the proposed occupant-centric control framework. To examine the system's behaviour under real-world conditions, on-site experiments were conducted in a test bed and a living lab environment. The primary objective of both experiments was to analyze the impact of system-level behaviour, driven by dynamic occupant states, on real-world operational environments, compared with conventional temperature-based control. The details of control methods are explained in the *Control methods* section.

Test-bed Setup. The test-bed was set up within a university building in Seoul, South Korea, and it measured 5.4 m × 2.0 m × 2.2 m, as shown in Figure 3(a). The test-bed features a packaged terminal heat pump system (Carrier CSV-Q075BI) for thermal environment control, with a cooling capacity of 2.8 kW, a heating capacity of 3.5 kW, a cooling COP of 3.64, and a heating COP of 3.89. In the test-bed environment, all PMV input variables were directly obtained without relying on assumed values. Environmental factors, including T_{air}, RH, T_{mrt} and V_{air}, were directly measured using IoT-based sensors. Meanwhile, personal factors, such as MET and CLO, were estimated in real time using the camera-based recognition algorithm described in the Occupant-centric Thermal Control Framework section. Real-time data collection was managed through Arduino Uno and Raspberry Pi boards, with a camera sensor utilized to capture occupant information.

Specifically, data collection relied on a dedicated hardware suite with specified accuracies. DHT11 sensors (accuracy: $T_{air} \pm 2$ °C, RH ±5%) measured T_{air} and RH. These were also utilized within a customized black bulb thermometer for T_{mrt} calculations. Additionally, V_{air} was monitored via a three-cup anemometer (accuracy: ±0.3 m/s), while building power usage was tracked by an ACS712ELC 30A T current meter (accuracy: ±1.5%). Occupant profiles were captured using a Raspberry Pi 8 MP camera board (2592 × 1944 pixels, 1080p at 30 fps).



Figure 3. Images of experimental environments: (a) test-bed and (b) living lab.

These specifications establish the baseline instrumental uncertainty for the subsequent thermal comfort evaluations. The test-bed was additionally equipped with a camera system for occupant image capture, power meters for electricity monitoring, and infrared sensors for automatic control. Data were initially collected at 15-s intervals and subsequently processed into 10-min averages.

The test-bed experiments were conducted by applying the two control strategies described in the *Control methods* section. All test-bed experiments were performed with a single occupant. A total of eight participants were recruited, with an average age of 26.5 years. To minimize inter-individual variability in thermal responses, participants were selected within a normal body mass index (BMI) range. The average BMI was 21.09 kg/m^2, with a standard deviation of 1.45 kg/m^2. This approach mitigated potential variations in thermal responses due to individual physical characteristics, thereby ensuring consistency in evaluating the control strategies. The experiments were conducted during the heating season.

The experimental scenarios consisted of nine conditions, defined by the cross-combination of three MET levels and three CLO levels. To reflect representative indoor activities, the MET levels were set to 1.0 met (sitting), 1.2 met (standing), and 1.7 met (walking), and participants performed the corresponding activities within the test-bed space. The CLO levels were set to 0.5, 0.75, and 1.0 clo to ensure the diversity of thermal resistance in the experiments. The base clothing ensemble consisted of underwear, ankle socks, and shoes, while additional clothing combinations were defined in accordance with ISO 7730.[16] For the 0.5 clo condition, participants wore a short-sleeve shirt and shorts; for the 0.75 clo condition, long pants and a long-sleeve shirt; and for the 1.0 clo condition, long pants, a long-sleeve shirt, and a sweater. Each of the eight participants completed all nine MET-CLO conditions, resulting in a total of 72 experimental sessions (8 participants × 9 conditions). Each session was defined as one participant performing a single MET-CLO condition for 40 min. During each session, the assigned activity level was maintained continuously throughout the entire period to represent the corresponding MET condition, while the assigned clothing ensemble remained unchanged to maintain the corresponding CLO condition.

Living Lab Setup. The living lab experiments were conducted in an actual office environment with multiple occupants to examine how the proposed occupant-centric control framework operates under real-world conditions. Unlike the controlled test-bed experiments, the living lab setting allows occupant behaviour, presence patterns, and environmental conditions to vary naturally over time.

The living lab was established in an operating research facility and was instrumented to continuously monitor occupant-related information and indoor environmental conditions, enabling real-time occupant-centric control. The living lab experiments were conducted over eight weekdays during the summer season, involving 11 occupants, with regular working hours from 10:00 AM to 6:00 PM. For comparative analysis, temperature-based control was employed. The indoor T_{air} was maintained at 24 °C in accordance with common practice during the cooling period.

The living lab environment consisted of an office space measuring 7.8 m × 8.6 m × 2.7 m, with a total area of approximately 67.1 m^2. As shown in Figure 3(b), cameras were strategically positioned to ensure comprehensive coverage of the entire space, thereby eliminating potential blind spots in occupant monitoring. The space was conditioned using ceiling-mounted electric heat pump (EHP) units configured in a split system. To minimize the direct influence of airflow on occupants, the airflow rate was set to a low level throughout the experiments, and wind deflectors were installed on each indoor unit. Similar to the test-bed setup, the living lab was equipped with IoT-based sensors to support control experiments, enabling continuous monitoring of indoor environmental conditions and energy consumption. In the living lab environment, T_{air} and RH were monitored using an AW1008KP sensor (Accuracy: T_{air} ± 0.3 °C, RH ±2.0%), providing

precise ambient measurements in the larger space. The camera sensors for occupant detection and the power measurement sensors remained identical to those used in the test-bed experiment. In addition, the data collection procedure followed the same protocol as the test-bed framework, with raw measurements collected at 15 s intervals and aggregated into 10 min averages for subsequent analysis.

Unlike the test-bed environment, where all PMV input variables were directly measured, the living lab presented practical constraints on comprehensive environmental sensing due to its larger, multi-occupant, and less controlled nature. Environmental sensors measured indoor T_{air} and RH, with four sensors installed at a height of 1.2 m and distributed across the office's central areas to ensure representative data collection. Direct measurement of T_{mrt} and V_{air} was not feasible in the living lab environment; therefore, these parameters were assumed using commonly adopted conventions in the literature. Specifically, T_{mrt} was assumed to be equal to indoor T_{air}, and V_{air} was set to 0.2 m/s based on typical office HVAC operating conditions.[17] Power sensors were used to monitor the EHP system's power consumption during operation. In an actual office environment, occupant recognition may be limited by partitions, desks, and monitors, which can obstruct the estimation of occupants' activities and clothing conditions.[29] To mitigate this limitation, real-time CLO estimation was performed using images captured in corridor areas, where occupants could be observed with minimal occlusion. To avoid repeated counting of the same individual, each occupant was assigned a unique ID, and the average CLO value across all identified occupants was used for control at each time step. MET was fixed at 1.0 met (seated activity), reflecting the predominant activity pattern in typical office environments.

Control methods

In this study, two control methods ($MODE_{OCC}$ and $MODE_{Temp}$) were employed to compare the effectiveness of occupant information on the system's operational carbon emissions. The occupant-centric control mode ($MODE_{OCC}$) uses PMV as a control parameter and incorporates estimated MET and CLO values to reflect dynamic occupant states in both simulation and field-test environments. The PMV setpoint for system control was set to 0, representing a thermally neutral state. Occupants were within the acceptable thermal comfort range when PMV values were between -0.5 and $+0.5$, consistent with the ASHRAE 55 criterion described above. No explicit PMV dead band was applied. Instead, at each control cycle, the control system selected the available discrete temperature setpoint (18–30 °C in 1 °C increments) that resulted in the PMV value closest to thermal neutrality (PMV = 0). The indoor temperature–based control ($MODE_{Temp}$), one of the representative control methods in buildings, served as the baseline method. While the building energy simulation assessed the sensitivity of operational carbon performance regarding dynamic occupant states, the on-site experiments were conducted to evaluate the $MODE_{OCC}$ against a baseline method ($MODE_{Temp}$).

Figure 4 conceptually illustrates the feedback control logic and key differences between the two control modes applied in this study. Both control modes follow a standard feedback control structure and use an identical 10-min control cycle. In $MODE_{Temp}$, indoor T_{air} is used as the control variable, and the error signal is defined as the difference between the temperature setpoint and the measured T_{air}. The system adjusts the indoor temperature accordingly in each control cycle.

In contrast, $MODE_{OCC}$ uses real-time PMV as the control variable, incorporating estimated occupant data (MET and CLO) along with environmental variables. Environmental data including T_{air}, RH, T_{mrt}, and V_{air}, as well as indoor images and system status (on/off, power consumption), are collected using on-site sensors. During each control cycle, PMV is calculated using the average values of the environmental factors over the 10-min interval and the representative (frequent) values of the estimated MET and CLO. This frequency-based representativeness approach applies consistently in both single- and multi-occupant environments, ensuring that the control system reflects the predominant occupant state in the space at each control cycle, even as individual occupants' states vary over time or differ among occupants present simultaneously. The calculated PMV is then compared with the setpoint (PMV = 0, neutral state), and the resulting error signal is used to adjust the indoor air temperature setpoint in the subsequent control cycle. Through this structure, dynamic changes in occupant state are continuously reflected in the control decision-making process.

Evaluation metrics

The performance of the occupant-centric thermal control system, which integrates dynamic occupant information, is quantitatively evaluated from two perspectives: thermal comfort and system-level carbon emissions. These metrics assess the extent to which occupant-state recognition yields measurable operational impacts in real-world building environments.

To evaluate building carbon performance, carbon emissions were calculated from the heating and cooling energy consumption associated with HVAC system operation for thermal environment control. This study quantitatively examines the impact of integrating real-time occupant states into HVAC control on carbon emissions, in comparison with a conventional control strategy. Carbon emissions during the building operation phase are primarily calculated based on energy consumption, using standardized estimation methods that apply emission factors specific to different energy sources. In

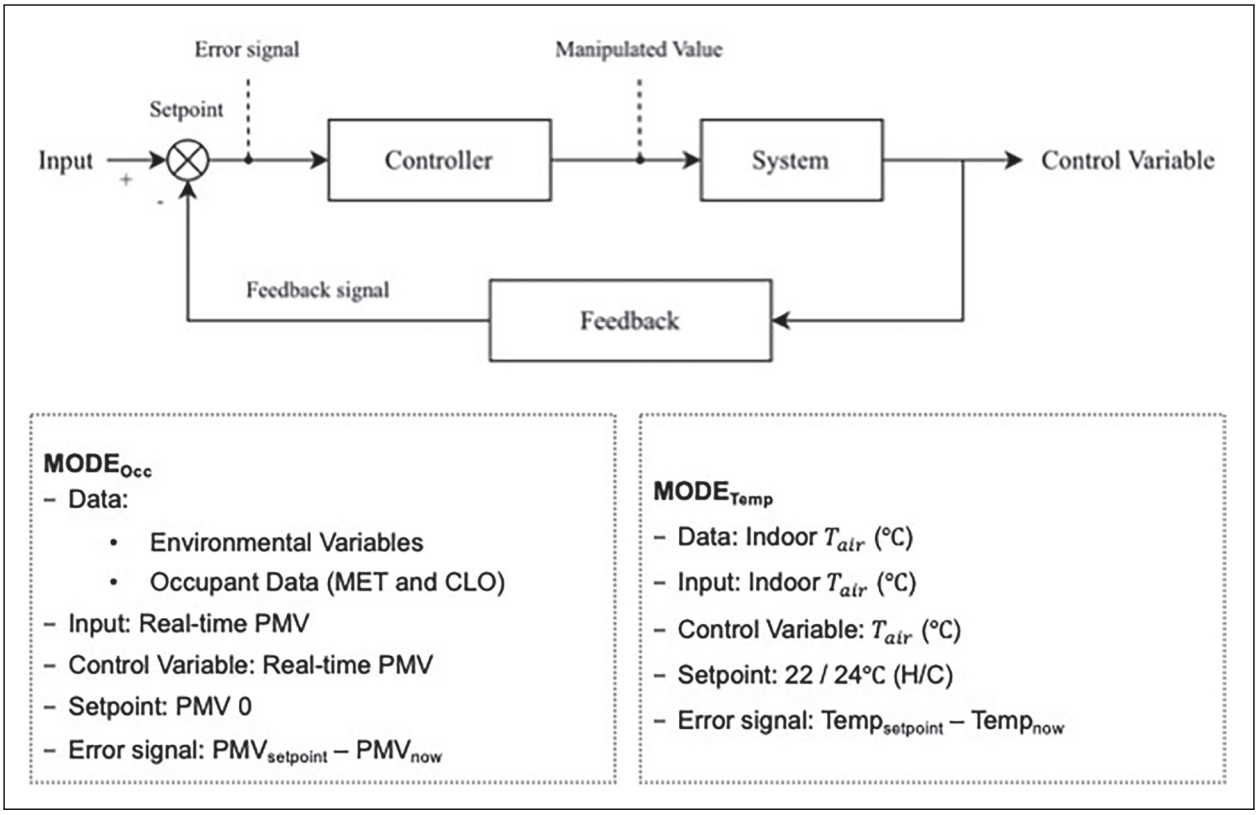


Figure 4. Feedback control logic and methods.

this study, greenhouse gas emissions generated during building operation are converted into carbon dioxide equivalents (CO_2eq), with carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) considered as the primary emission components. Global warming potential (GWP) values and energy-source–specific emission factors for electricity and gas are used in the calculation process.

Through this analysis, the extent to which changes in energy consumption induced by different building control strategies affect carbon emissions is evaluated. Importantly, the objective of this study is not limited to estimating absolute carbon emissions, but rather to assess how occupant-centric control strategies alter relative carbon-emission performance compared to a conventional temperature-based control. Detailed formulations for carbon emission calculations and the emission factors applied in this study are provided in Appendix 2.

Results

Simulation-based sensitivity analysis of carbon emissions to occupant states

This section presents a simulation-based analysis of how the proposed occupant-centric HVAC control responds to variations in occupants' personal factors. The analysis focuses on system-level carbon emission patterns associated with changes in MET and CLO. Unlike the real-time MET and CLO values obtained from the test-bed and living lab experiments, the values used in this simulation were systematically varied to evaluate sensitivity under controlled conditions rather than derived from real-time occupant measurements. Heating and cooling periods are examined separately to evaluate the resulting carbon emission characteristics. Simulation-derived energy consumption results were used to calculate carbon emissions during the building operation phase following the methodology described in the *Test-bed Validation Results* section and Appendix 2.

To clearly isolate the effects of occupant-centric HVAC control, the analysis considers only occupied periods during weekday working hours (08:00–19:00). Heating energy consumption is calculated as the combined energy use of the AHU heating coil and VAV reheat coils, while cooling energy consumption is defined as the energy use of the AHU cooling coil.

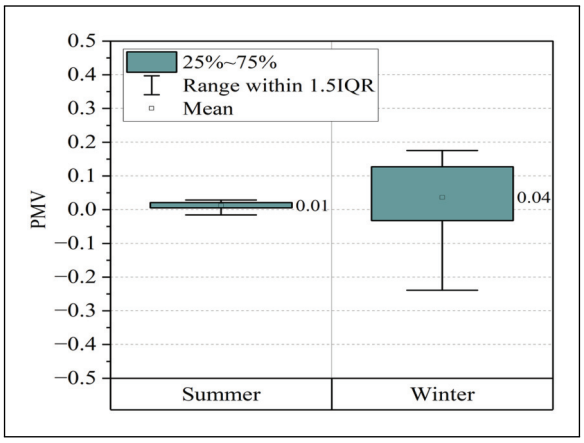


Figure 5. Distribution of simulated PMV values across MET–CLO combinations.

Across all simulation scenarios, the average PMV during occupied periods remained close to thermal neutrality, with mean values of 0.01 in summer and 0.04 in winter (Figure 5). PMV values during the cooling period remained close to thermal neutrality, ranging from −0.02 to 0.03, whereas those during the heating period exhibited a wider range (−0.24 to 0.18). Nevertheless, all simulation cases remained within the acceptable comfort range (−0.5 to +0.5) specified by ASHRAE 55, demonstrating that the proposed control strategy maintained thermal comfort despite variations in occupant states.

Figure 6 presents the daily average carbon emissions, expressed in $kgCO_2eq$, as a function of variations in MET and CLO. Each MET–CLO combination was held constant throughout the corresponding simulation run, and emission characteristics under heating and cooling conditions are visualized as heat maps. In the system configuration studied, heating is supplied by a combination of electricity and natural gas (LNG), whereas cooling relies exclusively on electricity. Accordingly, heating-related carbon emissions are further disaggregated by energy source. Figure 6(c) and 6(d) illustrate the respective contributions of electricity- and LNG-based emissions during the heating period. The analysis indicates that electricity-related emissions exhibit trends similar to the overall heating emission patterns shown in Figure 6(a), while LNG-related emissions remain at relatively low levels. While this behaviour stems from the VAV system's reheat-based heating mechanism, the observed carbon emission trends are potentially sensitive to the overall HVAC system architecture.

Overall, carbon emission patterns exhibit consistent trends across the evaluated conditions. During the heating period (Figure 6(a)), carbon emissions decrease as both MET and CLO increase, whereas an opposite trend is observed during the cooling period (Figure 6(b)). Notably, the influence of occupant-related factors on carbon emissions is more pronounced during heating than during cooling in this system. Heating-related emissions exhibit a wide variation range from 73.5 to 579.0 $kgCO_2eq$, while cooling-related emissions vary within a comparatively narrower range of 173.5 to 399.7 $kgCO_2eq$. This suggests a clear disparity in how occupant-related factors influence carbon outcomes between heating and cooling seasons.

This asymmetric behaviour is further illustrated through specific examples. Under winter conditions, reducing CLO from 1.0 to 0.5 clo at a seated MET (1.0 met) increases heating-related carbon emissions by approximately 126.8 $kgCO_2eq$. In contrast, a similar change in CLO under summer conditions results in a relatively minor change in emissions. Likewise, increasing activity level from sitting (1.0 met) to standing (1.2 met) at a CLO (0.75 clo) under winter conditions reduces heating-related carbon emissions by approximately 65.9 $kgCO_2eq$, while the corresponding increase in cooling-related emissions is approximately 13.7 $kgCO_2eq$. Overall, these findings – derived from a representative office model – suggest that integrating occupant state information into building control offers substantially greater potential for carbon mitigation during heating periods, as system operations are more sensitive to dynamic occupant variables in winter.

In addition to the absolute contributions shown in Figure 6(c) and 6(d), the relative share of electricity and natural gas in total heating-related energy use also varied depending on occupant state (MET and CLO) conditions. Across the simulated MET–CLO combinations, the electricity shares during the heating period shifted by up to approximately 10 percentage points. Under certain occupant conditions, heating was met entirely by electricity (100%), whereas under others, the electricity share decreased to approximately 90%, with the remainder supplied by natural gas. Rather than resulting in a uniform reduction across all energy sources, changes in occupant conditions altered the distribution of energy use between electricity and natural gas through their influence on HVAC operation. This variation in energy source composition can

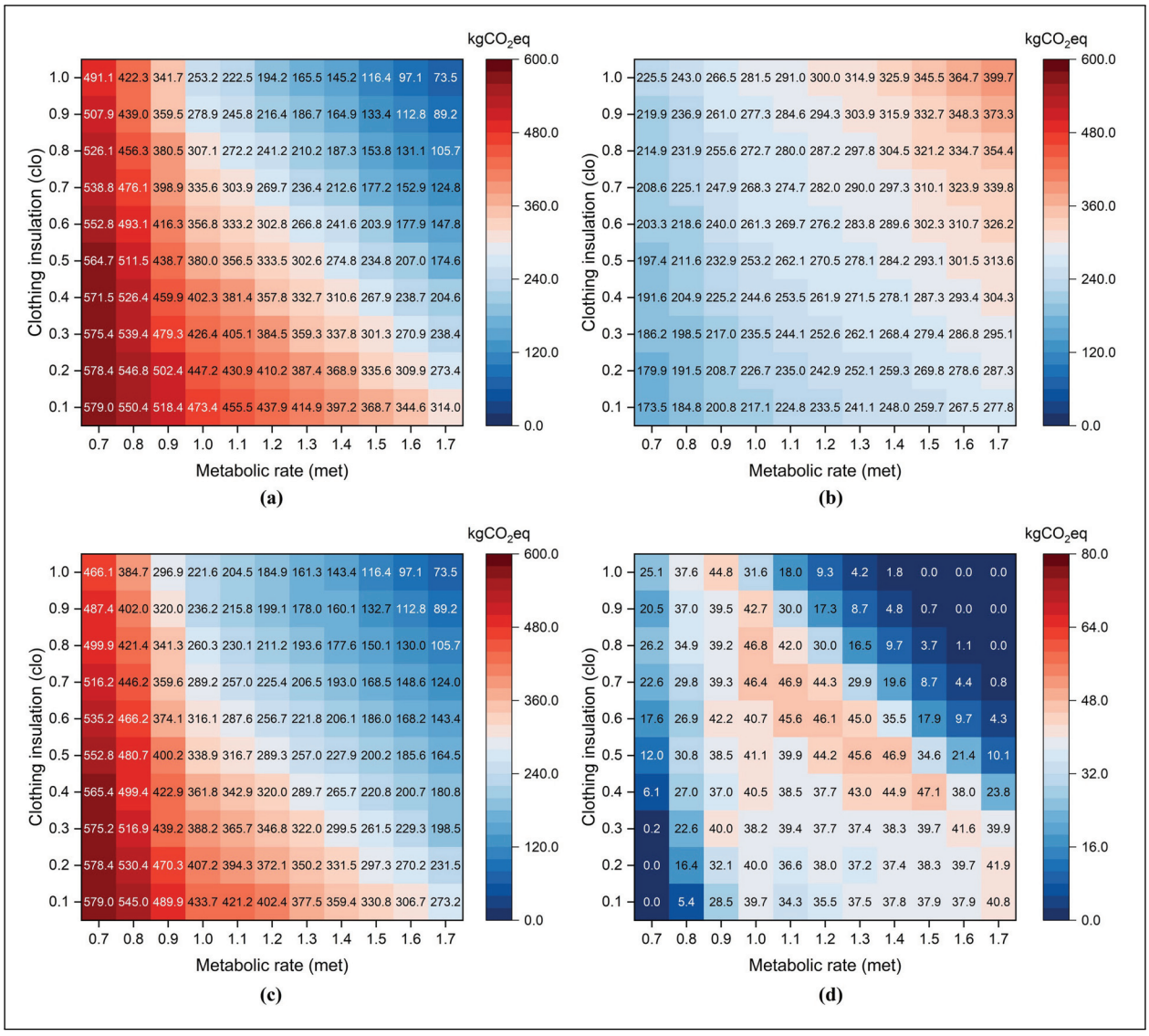


Figure 6. Carbon emission outcomes of occupant-centric control across varying occupant states: (a) total heating emissions, (b) total cooling emissions, (c) electricity-related heating, and (d) LNG-related heating.

consequently affect the associated carbon emissions, particularly when energy sources have different carbon intensities. Such effects may become more significant in buildings that rely on a broader and more diverse energy mix.

While the quantitative results of the simulation may vary with energy sources and system types, it is essential to analyze how transitions in occupant states influence overall carbon emissions patterns. To quantitatively evaluate this sensitivity, Figure 7 compares emission variations resulting from incremental changes of 0.1 units in MET and CLO. Figure 7(a) presents the average change in carbon emissions for a 0.1 clo increment, calculated by averaging the rates observed at fixed MET levels (ranging from 0.7 to 1.7 met). Similarly, Figure 7(b) illustrates the average emission response to a 0.1 met increase under fixed CLO conditions (ranging from 0.1 to 1.0 clo).

Consistent with the overall trends observed in Figure 6, carbon emissions increase with increasing MET and CLO under cooling conditions, whereas the opposite pattern is observed under heating conditions. However, the magnitude and variability of emission changes are substantially larger during heating. Specifically, a 0.1 clo increase results in a relatively stable increase of approximately 3–4% in cooling-related emissions across all MET levels, whereas heating-related emissions change by −2% to −14%, with larger reductions observed at higher MET levels. For instance, under the 1.0 met condition in Figure 7(a), a 0.1 clo increase leads to a 3% rise in cooling-related emissions, whereas heating-related

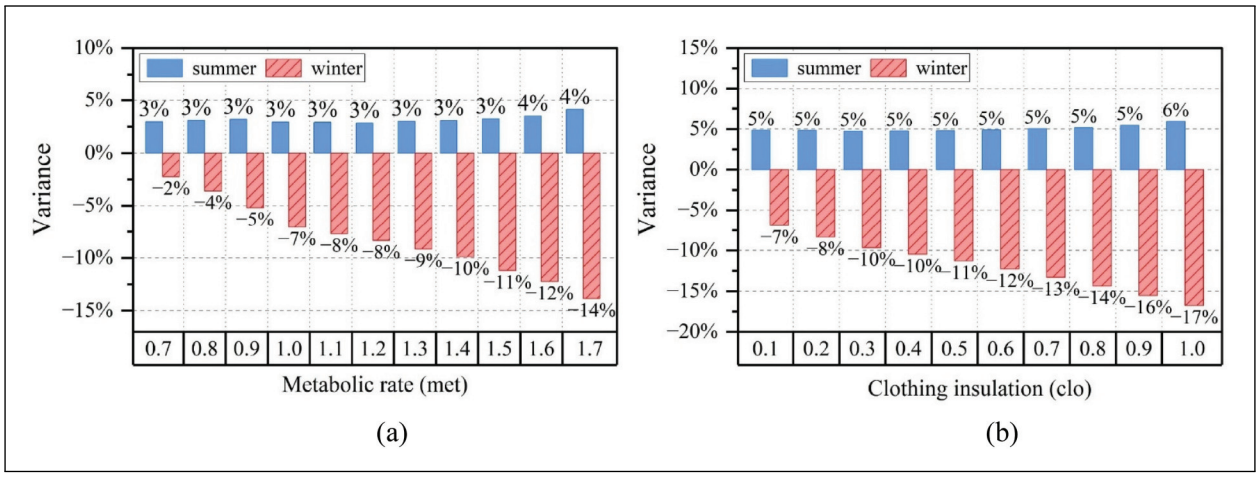


Figure 7. Sensitivity of carbon emissions to incremental changes in occupant-related factors: (a) 0.1 clo increase, (b) 0.1 met increase.

emissions decrease by 7%. In contrast, a 0.1-degree increase leads to approximately a 5–6% increase in cooling-related emissions, while heating-related emissions change by −7% to −17%.

This further confirms that occupant-driven adjustments have a more pronounced impact during the heating season than the cooling season. Specifically, the data reveal that MET variations are a stronger driver of emission shifts than CLO variations for an equivalent change in units. A direct comparison of the two cases shows that MET changes produce approximately 2% greater emission responses than CLO changes under cooling conditions and approximately 4–5% greater responses under heating conditions. Consequently, occupant MET exerts a stronger influence on carbon emission variability than CLO, with this sensitivity most pronounced during the heating period.

These results demonstrate that even small changes in occupant state, when explicitly reflected in control logic, can systematically alter system-level carbon emission outcomes. Consequently, control strategies based on human state recognition transition from being purely comfort-oriented to serving as interactive mechanisms that actively modulate carbon performance.

Overall, the simulation results presented in this section demonstrate that the proposed occupant-centric control leads to differentiated building system operation in response to changes in occupants' activity levels and clothing conditions. Among the variables examined, MET emerges as the dominant driver of carbon performance compared with CLO. The unit-based sensitivity analysis further demonstrates that incorporating such subtle variations enables more adaptive and responsive building operations, transcending the limits of conventional comfort maintenance.

Test-bed validation results

This section presents the results of test-bed experiments conducted to examine whether the simulation-based findings can be reproduced in a real physical system environment. In particular, the analysis focuses on how the occupant-centric HVAC control strategy ($MODE_{OCC}$) adjusts control behaviour in response to occupants' MET and CLO, and how it differs from conventional temperature-based control ($MODE_{Temp}$) in terms of thermal comfort and operational carbon emissions.

Each experimental condition lasted 40 min. During the initial 10 min, the indoor air temperature was maintained at 22 °C across all conditions to ensure consistent initial states. Thermal comfort and carbon emissions were then analyzed using data collected during the subsequent 30-min period after control activation. The experiments involved a total of eight participants, and the results were evaluated using averaged values across all participants.

During the experiments, $MODE_{OCC}$ dynamically adjusted the indoor T_{air} within an approximate range of 20–25 °C, depending on occupants' MET and CLO conditions, whereas $MODE_{Temp}$ maintained a constant indoor T_{air} of 22 °C for all conditions. To determine if occupant-centric control was properly implemented, Figure 8 compares thermal comfort outcomes across nine MET–CLO combinations to assess the PMV levels under varying occupant conditions. Overall, $MODE_{OCC}$ exhibited adaptive control behaviour by adjusting control outputs in response to changes in occupant states and maintained PMV values close to thermal neutrality (PMV = 0). Compared to $MODE_{Temp}$, $MODE_{OCC}$ remained largely within the acceptable comfort range, providing a more comfortable indoor environment.

Under certain conditions (1.0 met with 0.5 and 0.75 clo) in Figure 8, which correspond to relatively high heating demand, transient responses were observed because the initial indoor temperatures were lower than the target levels. As

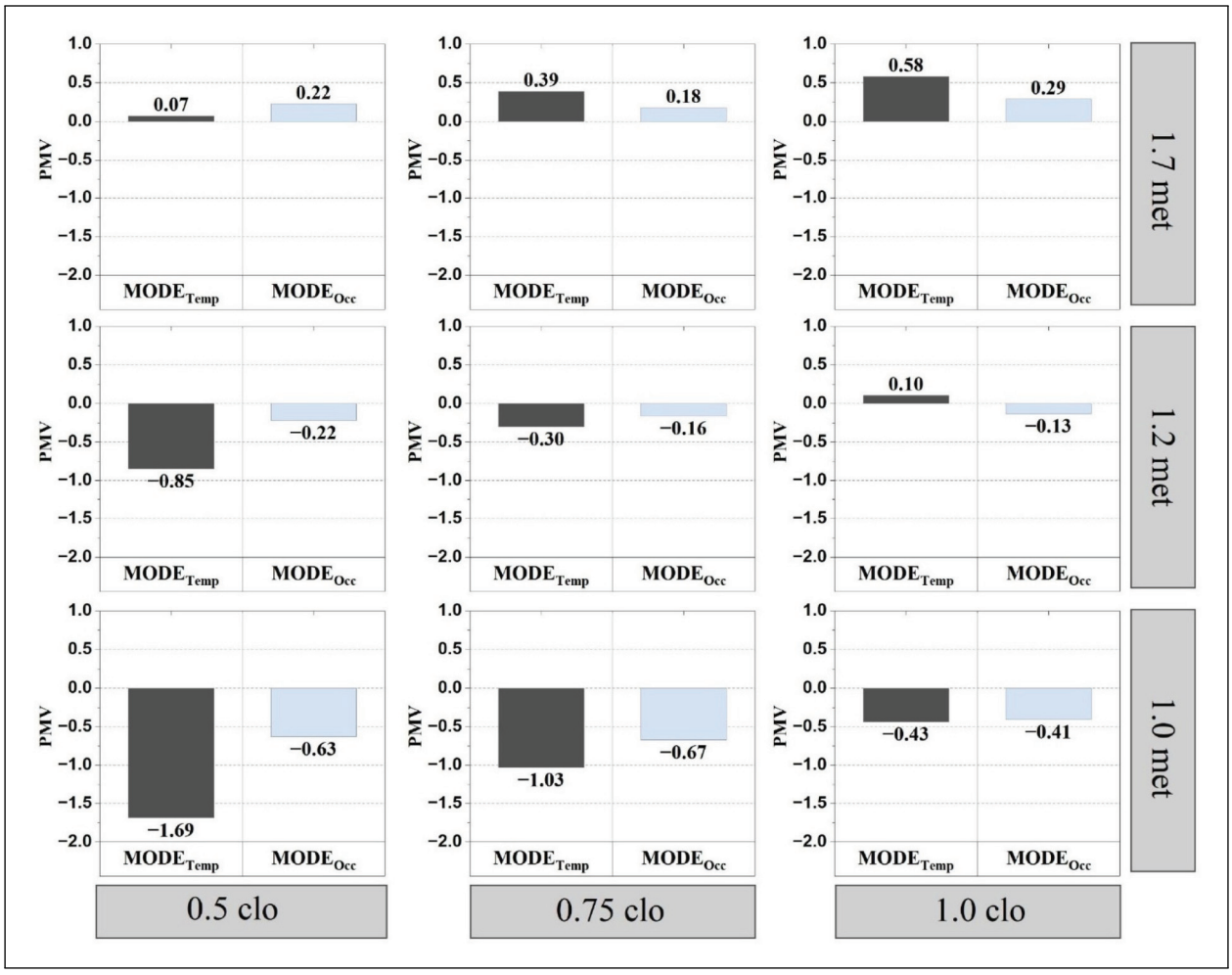


Figure 8. Thermal comfort results of the test-bed experiments.

a result, longer stabilization times were required for the control response, leading to extended periods of discomfort and slightly lower average thermal comfort values under $MODE_{OCC}$. Nevertheless, excluding these initial transient periods, $MODE_{OCC}$ demonstrated more adaptive responses to occupant states.

Figure 9 presents the carbon emissions from the test-bed experiments, expressed in $kgCO_2eq/h$. Figure 9(a) shows the results under $MODE_{Temp}$, which does not incorporate occupants' MET and CLO as control inputs and maintains a constant indoor temperature setpoint regardless of occupant states. The same MET–CLO-based heatmap format was retained to facilitate direct comparison with Figure 9(b). Under $MODE_{Temp}$ (Figure 9(a)), carbon emissions remained relatively constant regardless of variations in occupants' MET and CLO conditions, ranging from 0.80 to 1.0 $kgCO_2eq/h$ across the nine scenarios (average of approximately 0.86 $kgCO_2eq/h$). This result reflects that temperature-based control does not incorporate occupant state information into its decision-making process. Between consecutive sessions, the indoor environment was reset to a baseline condition before the subsequent experimental condition began. However, under periods of extreme outdoor temperature, the transition period may not have been sufficient to fully eliminate the thermal influence of the previous session. Consequently, residual thermal effects could have contributed to the observed variation in carbon emissions. Combined with the initial transient response discussed in the Test-bed Validation Results section, these effects may explain the isolated deviation observed in the $MODE_{Temp}$ carbon emission results (Figure 9(a)), despite this control mode being independent of occupants' MET and CLO conditions.

In contrast, the results obtained under $MODE_{OCC}$ (Figure 9(b)) show that heating control behaviour varies with occupant state conditions, leading to meaningful differences in carbon emissions. Emission levels range from 0.61 to 1.31 $kgCO_2eq/h$ across the evaluated conditions, demonstrating that control outcomes can differ substantially depending on occupant states even under the same indoor environment. For example, under low MET and low CLO conditions,

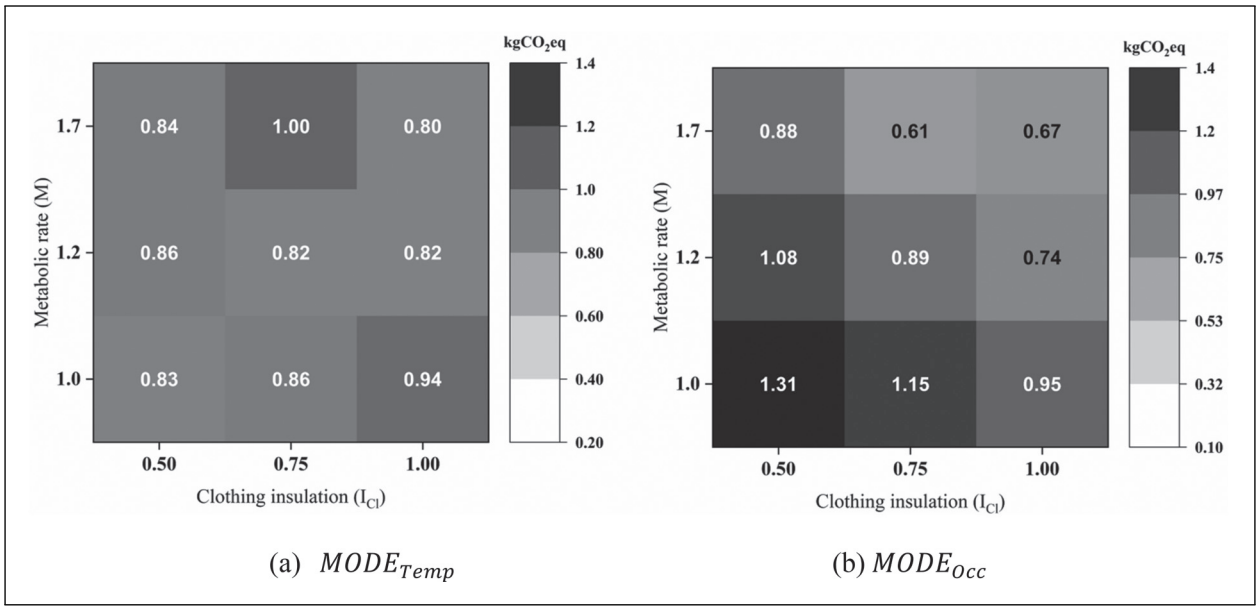


Figure 9. Carbon emission responses under different control modes in test-bed experiments. (a) $MODE_{Temp}$ and (b) $MODE_{Occ}$.

increased heating demand results in higher carbon emissions compared to $MODE_{Temp}$. Conversely, under conditions of high MET (1.7 met) combined with medium to high CLO levels, the system assesses a reduced need for heating and adjusts the indoor temperature setpoint to approximately 19–21 °C, thereby minimizing heating operation. As a result, the system remains largely in a standby state under these conditions and exhibits similarly low emission levels.

These test-bed results empirically confirm, consistent with the simulation-based analysis, that integrating occupant information into control logic leads to variations in control behaviour, thermal comfort, and carbon emissions in real-world system environments. In other words, occupant-centric HVAC control operates not merely as a setpoint-maintaining strategy, but as an adaptive and interactive building control mechanism that responds to changes in occupant states.

Living lab validation results

This section analyzes the system behaviour of $MODE_{OCC}$ in a real office living lab environment characterized by multiple occupants and sensing constraints, in comparison with $MODE_{Temp}$. The purpose of this experiment is to examine whether the occupant state sensitivity observed in the test-bed experiments remains valid under realistic operational conditions.

The living lab experiment was conducted by applying each control mode for four consecutive days. During the cooling period, the indoor temperature setpoint under $MODE_{Temp}$ was fixed at 24 °C.[28] Figure 10 illustrates the average occupancy patterns over the 8-day living lab experiment. The analysis was limited to official working hours (10:00–18:00). Within these hours, any periods identified as unoccupied, including lunch breaks, were excluded from the dataset. Under $MODE_{Temp}$, the average and maximum number of occupants were 6.5 and 9.8, respectively, while $MODE_{OCC}$ showed an average of 5.8 and a maximum of 9.6, demonstrating overall similar occupancy trends.

Figure 11 presents the indoor T_{air} and PMV results obtained from the living lab experiment. During working hours, $MODE_{Temp}$ maintained the indoor T_{air} within a range of 24.0–24.9 °C, with an average of 24.5 °C. In contrast, $MODE_{OCC}$ adjusted the indoor environment to a higher temperature range of 26.0–27.7 °C, with an average of 26.7 °C, by reflecting occupants' state information (Figure 11(a)). This result indicates that incorporating occupant state variables into control decisions enables higher temperature setpoints than conventional temperature-based control.

These differences in control behaviour are clearly reflected in thermal comfort outcomes (Figure 11(b)). Under $MODE_{Temp}$, the average PMV was −0.72, with approximately 50% of the data distributed between −0.47 and −0.94, indicating a tendency to fall outside the acceptable comfort range. In contrast, $MODE_{OCC}$ resulted in an average PMV of 0.30, with most values distributed within the range of 0 to 0.5, providing a more comfortable indoor environment for occupants. This finding demonstrates that $MODE_{OCC}$ resets control reference points based on occupants' thermal states and reflects human–building interaction in real-world operation.

Figure 12(a) displays the average HVAC carbon emissions during the living lab experiment, and Figure 12(b) compares the average emissions of each control method over four days. Under $MODE_{Temp}$, hourly carbon emissions peaked

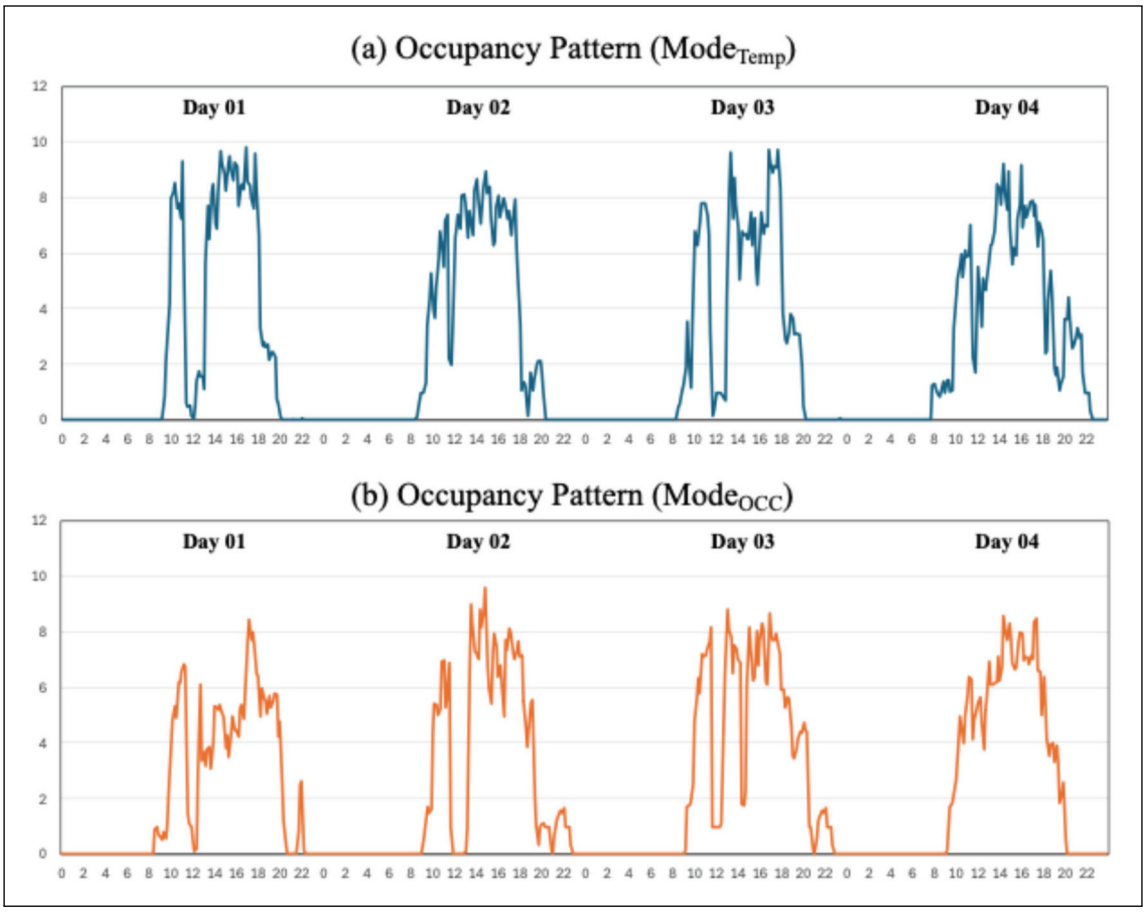


Figure 10. Occupancy patterns during the living lab experiment.

at up to 0.46 kgCO_2eq during the afternoon period (14:00–16:00), and the average daily cumulative emissions reached 18.77 kgCO_2eq. This pattern directly reflects the increase in cooling loads caused by rising outdoor air temperatures throughout the day.

In contrast, $MODE_{OCC}$ maintained indoor temperature setpoints approximately 2 °C higher than those under $MODE_{Temp}$ while keeping PMV values within the acceptable comfort range. As a result, the average daily cumulative carbon emissions under $MODE_{OCC}$ were reduced to 6.95 kgCO_2eq, corresponding to an approximate 63.0% reduction compared to $MODE_{Temp}$. Under $MODE_{OCC}$, greater variability in real-time emission levels was observed due to occupant states, leading to more active control decisions.

Because the observed carbon emissions could have been influenced by differences in outdoor temperature, an additional analysis was performed using two representative days with similar outdoor conditions. Although the experiments were conducted one week apart in September, the average outdoor temperature during the $MODE_{Temp}$ period (32.8 ± 3.2 °C) was approximately 5 °C higher than that during the $MODE_{OCC}$ period (27.0 ± 2.8 °C). To minimize this effect, Sep. 8 ($MODE_{Temp}$, average outdoor temperature: 31.5 °C) and Sep. 12 ($MODE_{OCC}$, average outdoor temperature: 31.0 °C) were selected for comparison, resulting in an average outdoor temperature difference of only 0.5 °C.

Figure 13 compares the indoor air temperature, temperature setpoint, and PMV obtained under the two control modes. Under $MODE_{Temp}$ (Figure 13(a)), the setpoint remained constant at 24 °C, resulting in a nearly constant indoor T_{air} with an average of 24.5 °C. Consequently, the average PMV was −0.78 and reached a minimum of −1.51, while only 23% of the occupied period remained within the ASHRAE 55 comfort range.

In contrast, under $MODE_{OCC}$ (Figure 13(b)), the temperature setpoint dynamically varied between 24 and 28 °C, resulting in indoor T_{air} ranging from 25.2 to 27.7 °C, with an average of 26.9 °C. The average PMV was maintained at 0.42, and the proportion of occupied time within the acceptable comfort range increased to 54.2%, more than twice that observed under $MODE_{Temp}$. Moreover, PMV deviations outside the comfort range were substantially smaller than those under conventional $MODE_{Temp}$.

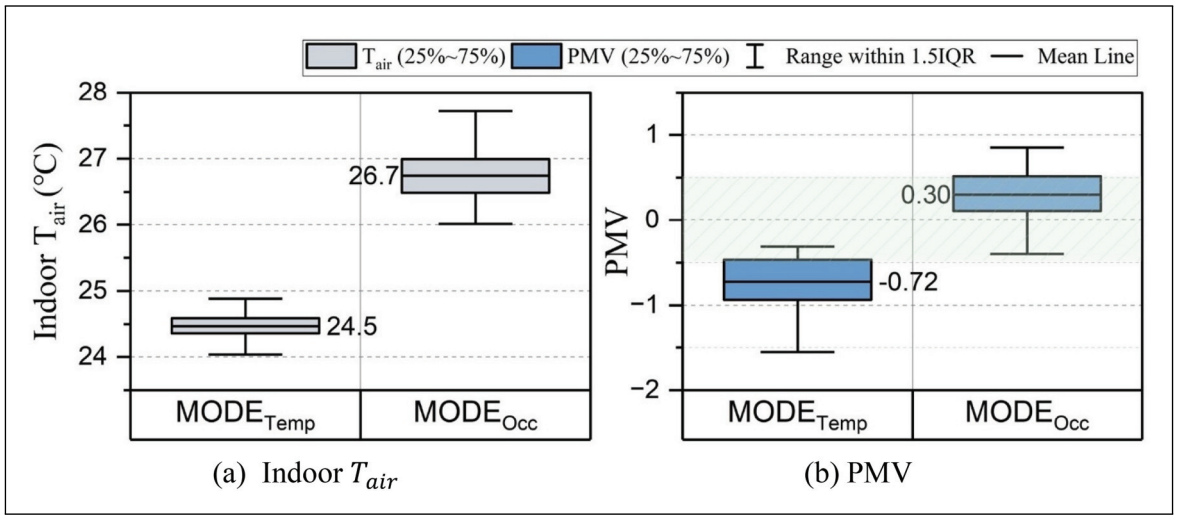


Figure 11. Control results of the living lab experiments. (a) Indoor T_{air} (b) PMV.

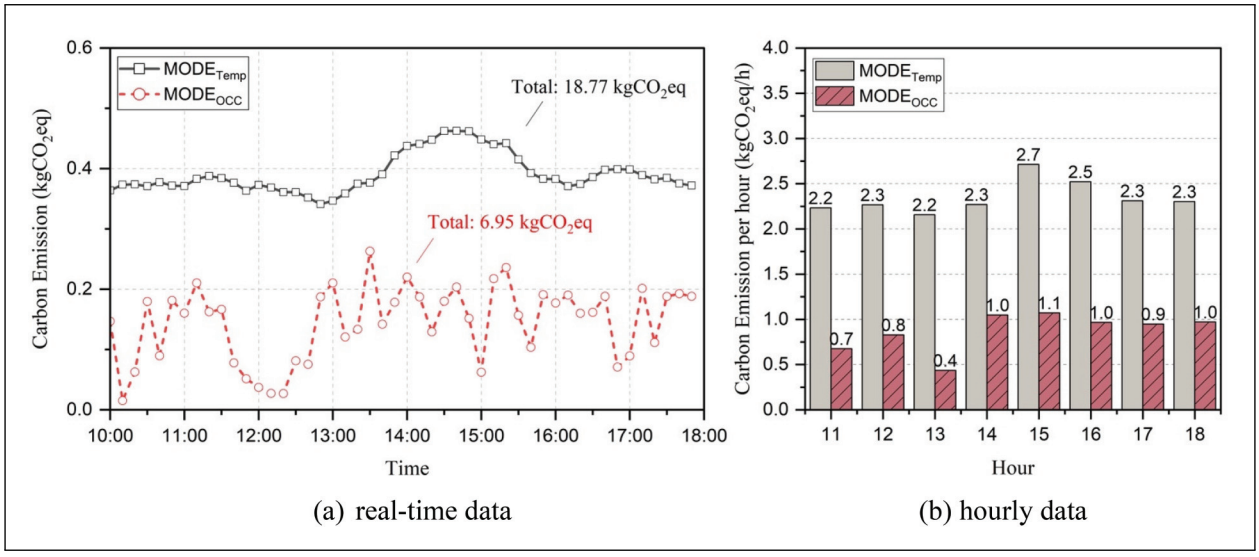


Figure 12. Average carbon emissions of the HVAC system by control mode. (a) Real-time data and (b) hourly data.

The resulting carbon emissions under each control mode are presented in Figure 14. Even with the influence of outdoor temperature minimized, $MODE_{OCC}$ exhibited lower carbon emissions than $MODE_{Temp}$ for most of the day. In terms of total daily carbon emissions, $MODE_{OCC}$ recorded 9.45 $kgCO_2eq$, corresponding to a 42.8% reduction relative to the 16.53 $kgCO_2eq$ observed under $MODE_{Temp}$. These results indicate that, even after minimizing the influence of outdoor temperature, control based on accurately measured occupant thermal comfort remains more effective in reducing carbon emissions under real-world conditions.

These living lab results demonstrate that control strategies that integrate human state information can achieve meaningful reductions in building operational carbon emissions while improving thermal comfort, even in environments with multiple occupants and realistic operational constraints. Although a relatively high level of carbon emissions reduction was observed, the magnitude of this reduction may vary by building type and operating conditions. The present living lab experiment was conducted in an office environment, where occupants primarily engage in sedentary activities, allowing greater thermal tolerance for higher indoor setpoints during the cooling period. In addition, the level of reduction may depend on the baseline setpoint adopted in the $MODE_{Temp}$ scenario used for comparison. Therefore, rather than generalizing the absolute reduction values, this study's significance lies in clearly demonstrating the operational impact of integrating dynamic occupant information into building control.

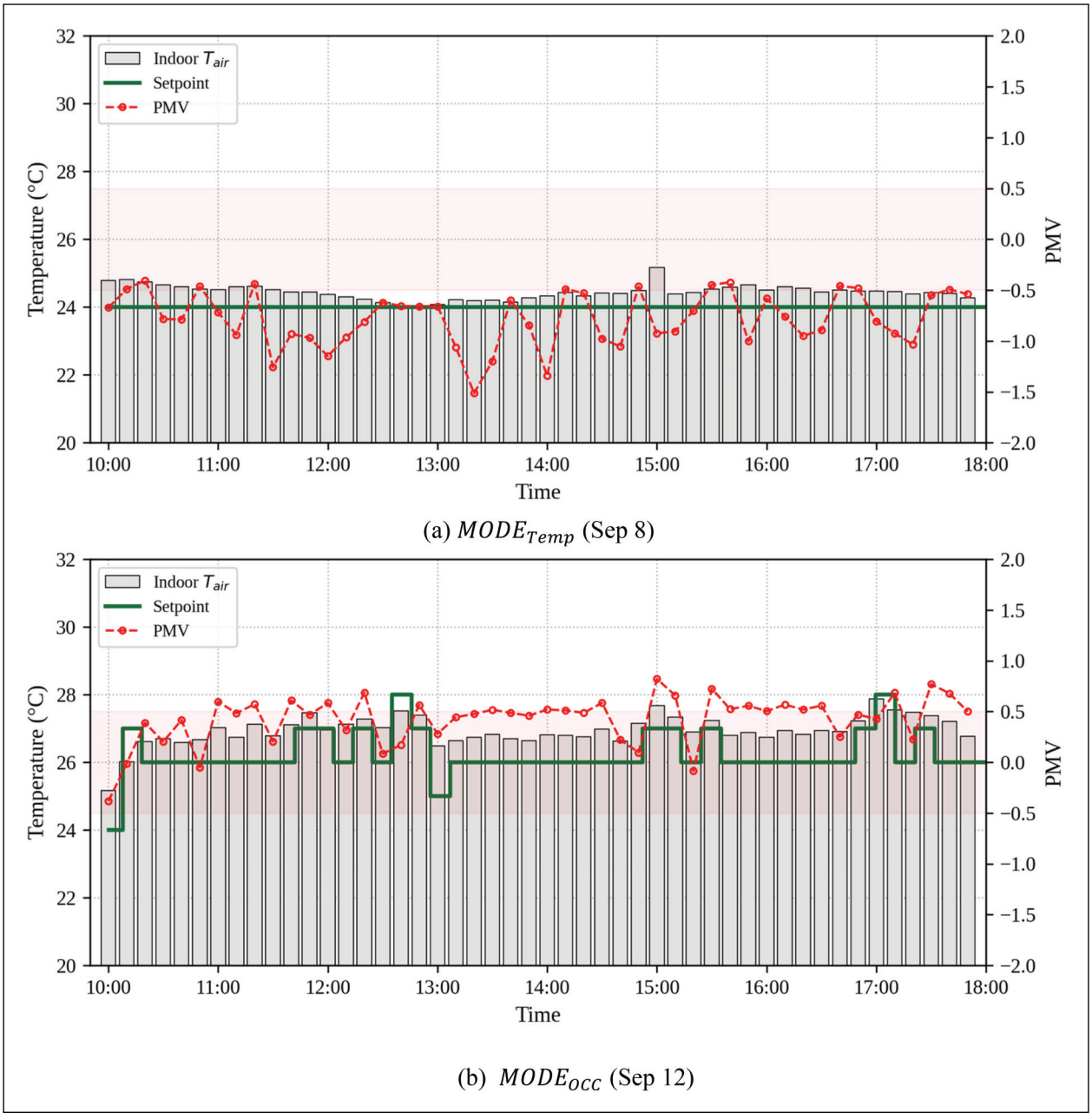


Figure 13. Indoor temperature and PMV variation by control method under similar outdoor temperature conditions. (a) $MODE_{Temp}$ (Sep 8) and (b) $MODE_{OCC}$ (Sep 12).

Discussion

The main theoretical and practical contributions of this study are as follows. While our prior works[29,33,39] established the feasibility of real-time MET/CLO estimation and its effect on thermal comfort–based control, this study explicitly redefines the outcome metric as operational carbon performance and newly isolates the relative contribution of MET and CLO to carbon emissions. First of all, this study quantitatively demonstrates that dynamic occupant variables (MET and CLO) are not merely thermal comfort adjustment factors, but critical control inputs that can fundamentally influence operational carbon performance in buildings. Both simulation and on-site experimental results show that variations in occupant activity and clothing levels produce systematic sensitivities in HVAC operation patterns and carbon emissions, with more pronounced effects observed under heating conditions. These findings suggest that conventional control strategies, or those based on static occupant assumptions, may have inherent limitations for carbon optimization.

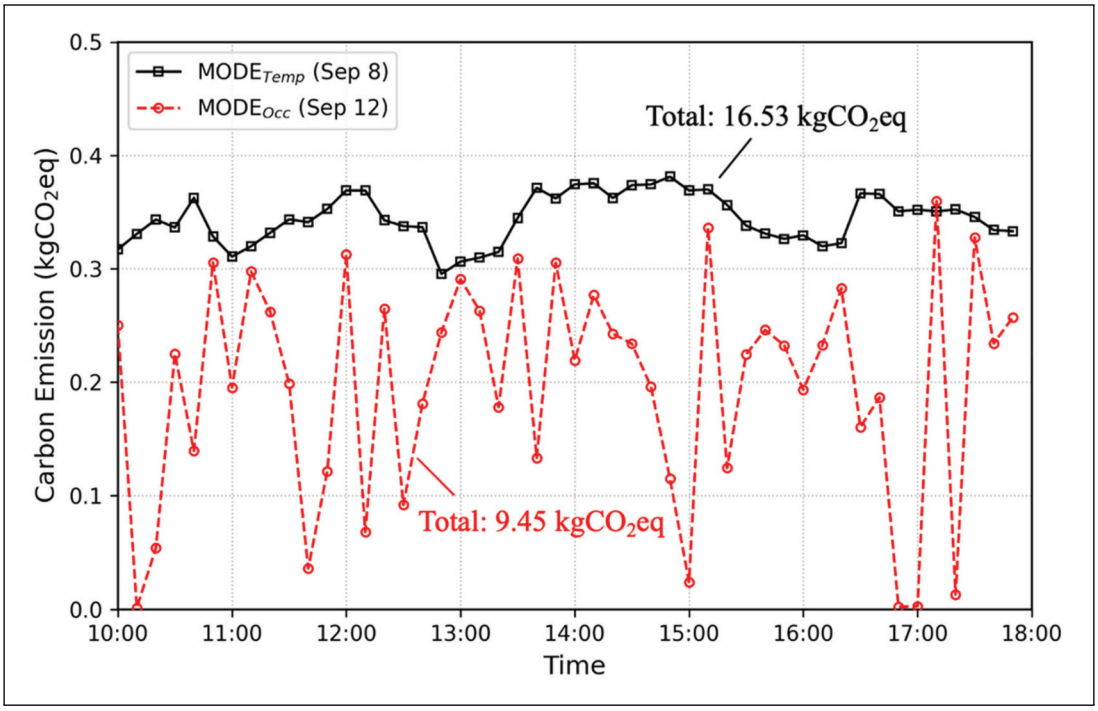


Figure 14. Comparison of carbon emissions over a single-day period under similar outdoor temperature conditions (Sep 8 & 12).

The results further indicate the need to extend HVAC systems beyond passive control structures that maintain fixed setpoints, toward interactive control systems that adjust decision-making in response to changes in occupant states. Accordingly, occupant-centric control should be considered a necessary component of future building operation, supporting multi-objective approaches that simultaneously account for thermal comfort and carbon performance.

Furthermore, beyond simulation-based analysis, this study validates the practical applicability of occupant-centric control through both controlled test-bed experiments and living lab studies. The fact that meaningful carbon reductions were observed despite the complexities of real-world office constraints, such as partitions, furniture layouts, and varying occupant presence, demonstrates that the proposed approach is not merely theoretical.

Uncertainty analysis

Because all environmental variables were measured using physical sensors, the uncertainty associated with sensor measurements was evaluated to assess the reliability of the reported results. Uncertainty analysis of PMV and carbon emission estimates was conducted for both the test-bed and the living lab using manufacturer-specified sensor accuracies.

In the test-bed, all environmental variables (T_{air}, RH, T_{mrt}, and V_{air}) were directly measured. The uncertainty analysis was therefore performed using Monte Carlo simulation. Monte Carlo simulation was performed by randomly sampling uncertain input variables according to their assumed distributions. This process was repeated 20,000 times to characterize the resulting output distributions. Sensor measurement uncertainty was evaluated following the Guide to the Expression of Uncertainty in Measurement (JCGM 100:2008).[40] Since only the manufacturer-specified accuracy ranges were available in this study, sensor values were assumed to follow a uniform distribution within their specified accuracy range, consistent with a Type B (non-statistical) evaluation.

The Monte Carlo simulation was performed for each of the nine MET and CLO conditions in the test-bed, and the results are shown in Table 2. The setpoint uncertainty shown in Table 2 was derived as follows. First, the 95% uncertainty interval of PMV ($PMV_{95\%UI}$) represents the variation in PMV arising from sensor measurement uncertainty. Second, $\Delta PMV/\Delta T$ indicates how much PMV changes per one degree change in temperature under that condition. Dividing $PMV_{95\%UI}$ by $\Delta PMV/\Delta T$ yields the corresponding setpoint uncertainty ($U_{setpoint}$, ±°C), representing the degree to which the control system may select a setpoint that deviates from the true optimum due to this uncertainty.

The resulting $PMV_{95\%UI}$ ranged from 0.33 to 1.01 across the nine conditions. For each condition, $\Delta PMV/\Delta T$ was calculated using the measured mean environmental conditions in $MODE_{OCC}$ as the baseline. The resulting $U_{setpoint}$ ranged from ±2.2 to ±2.8 °C across the nine conditions, with an average of 2.47 °C. Although $PMV_{95\%UI}$ varied by more than

Table 2. PMV and setpoint uncertainty according to MET and CLO conditions.

MET	CLO	PMV95%UI	ΔPMV/ΔT	$U_{setpoint}$ (±°C)
1.0	0.5	1.01	0.38	2.7
1.0	0.75	0.87	0.31	2.8
1.0	1.0	0.68	0.25	2.7
1.2	0.5	0.72	0.30	2.4
1.2	0.75	0.59	0.25	2.4
1.2	1.0	0.50	0.21	2.4
1.7	0.5	0.50	0.20	2.3
1.7	0.75	0.41	0.18	2.3
1.7	1.0	0.33	0.15	2.2

threefold across conditions, $U_{setpoint}$ remained relatively consistent (±2.2 to 2.8 °C). At lower MET and CLO values, both PMV uncertainty and temperature sensitivity (ΔPMV/ΔT) increased simultaneously. Because these effects offset each other, the resulting $U_{setpoint}$ remained similar across all conditions.

The effect of this setpoint uncertainty on carbon emissions was then estimated using Equations (1) and (2). Carbon emission sensitivity to temperature ($S_{CO_2eq/T}$) was calculated from the observed carbon emission range (0.61 to 1.31 $kgCO_2eq/h$) and the corresponding setpoint range (20 to 25 °C), as defined in Equation (1). Multiplying $S_{CO_2eq/T}$ by the average $U_{setpoint}$ (2.47 °C) yielded the uncertainty in carbon emissions (U_{CO_2eq}), as defined in Equation (2). The resulting carbon emission uncertainty was ±0.35 $kgCO_2eq/h$.

$$S_{CO_2eq/T} = \frac{\Delta CO_2eq}{\Delta T} \tag{1}$$

$$U_{CO_2eq} = U_{setpoint} \times S_{CO_2eq/T} \tag{2}$$

Where $S_{CO_2eq/T}$ is carbon emission sensitivity to temperature ($kgCO_2eq/h/°C$), $U_{setpoint}$ is setpoint uncertainty (°C), U_{CO_2eq} is uncertainty in carbon emissions ($kgCO_2eq/h$).

In the living lab, only T_{air} and RH were considered sources of instrumental uncertainty. Because daily averages were calculated from 48 measurements collected at 10-min intervals, random measurement error was substantially reduced, and the resulting $PMV_{95\%UI}$ was ±0.03, considerably smaller than in the test-bed. This reflects both the higher measurement accuracy of the AW1008KP sensor (±0.3 °C) and the error-reduction effect of daily averaging.

Applying the same approach used for the test-bed, the ΔPMV/ΔT at the representative living lab condition was 0.46, yielding a $U_{setpoint}$ of 0.07 °C. Carbon emissions ranged from 0.19 to 1.84 $kgCO_2eq/h$, while indoor temperature varied from 26.62 to 27.47 °C, yielding a temperature sensitivity of 1.95 $kgCO_2eq/h/°C$. Multiplying this value by $U_{setpoint}$ (±0.07 °C) resulted in a carbon emission uncertainty of ±0.13 $kgCO_2eq/h$.

Overall, the effect of instrumental uncertainty on carbon emissions was larger in the test-bed (±0.35 $kgCO_2eq/h$) than in the living lab (±0.13 $kgCO_2eq/h$). This difference primarily reflects the lower measurement accuracy of the DHT11 sensor used in the test-bed compared with the AW1008KP sensor used in the living lab. The estimated setpoint uncertainty represents the theoretical propagation of sensor measurement uncertainty based on manufacturer-specified accuracy ranges under a Type B evaluation, rather than the actual control error of the proposed system. In the experiments, measurements from multiple sensors were averaged, and 10 min averaged environmental data were used for control decisions, further reducing the influence of random measurement fluctuations. Nevertheless, sensor measurement uncertainty can still propagate through PMV estimation and influence setpoint selection and the resulting carbon emission estimates. Therefore, future studies should investigate uncertainty-aware control strategies and incorporate higher-accuracy sensors and sensor calibration procedures to further improve the reliability of occupant-centric thermal control.

Limitations and future work

Nevertheless, several limitations should be acknowledged. The three-stage framework was designed to progress from controlled to real-world conditions. The test-bed isolated the causal effects of MET and CLO under fixed experimental conditions, whereas the living lab relaxed these experimental controls, allowing occupant presence, occupant behaviour, and environmental conditions to vary naturally, as reflected in the observed occupancy patterns (Figure 10). Nonetheless, because the living lab was conducted in a single research office, it does not capture the full diversity of occupant behaviours and operational constraints encountered across a wider range of commercial buildings, such as varying tenant compositions and flexible work arrangements.

In addition, the absolute magnitude of carbon emissions and reduction rates may vary depending on building scale and system configurations. This study was limited to a specific office environment and test-bed setting and therefore did not fully capture the diversity of building types, climate conditions, energy sources, or occupancy patterns encountered in real-world applications. While future long-term validation across a wider range of scenarios is needed to strengthen the generalizability of the findings, the proposed control framework is based on the physical relationships embedded in the PMV model. Consequently, although the magnitude of carbon emission reduction is expected to remain context-specific, the qualitative trends identified in this study, such as the greater influence of MET than CLO and the higher sensitivity during the heating season, are expected to remain consistent across different building types and climate conditions. Furthermore, the simulation-based sensitivity analysis was conducted using the standardized ASHRAE 90.1 medium office reference model, providing broader applicability than would be achieved using a single living lab alone.

Furthermore, carbon emissions were estimated using annual average national grid emission factors published by the Greenhouse Gas Inventory and Research Center of Korea (GIR); as these factors do not capture intra-day or seasonal variability in the generation mix (e.g., fluctuating renewable energy shares), the estimated carbon reduction may differ under real-time or forecasted grid conditions. However, because both control modes were evaluated under the same emission-factor framework, this limitation is expected to have minimal impact on the relative comparison between $MODE_{Temp}$ and $MODE_{OCC}$ presented in this study.

The PMV model was originally developed for steady-state conditions and may therefore have limited applicability under transient thermal conditions. This limitation is particularly relevant during the initial stabilization period of the test-bed experiments, when indoor temperatures had not yet fully reached their target levels. Although the use of 10 min interval averaged measurements partially mitigates the influence of short-term fluctuations, transient thermal responses are not explicitly captured by the PMV model. Future work should therefore investigate transient or adaptive thermal comfort models to further improve the robustness of occupant-centric thermal control under dynamically changing environmental conditions.

While all PMV input variables were directly measured or estimated in the controlled test-bed environment, the living lab experiment required fixed values for T_{mrt}, V_{air} and MET, as direct measurement of these variables was not feasible in the larger, multi-occupant, and less controlled office setting. Although such assumptions are commonly adopted in PMV-based studies under practical constraints, they may introduce discrepancies between estimated and actual PMV values, particularly under conditions involving strong radiant asymmetry (e.g., near windows) or variable airflow (e.g., near HVAC diffusers or open doors).[33] Therefore, the absolute PMV values obtained from the living lab experiment should be interpreted with consideration of these uncertainties. However, because the same assumptions were applied consistently across both control modes, their influence on the relative comparison between the two strategies is expected to be limited. Future studies should incorporate more accurate, occupant location-aware environmental measurements to improve PMV estimation under real-world operating conditions. To better distinguish between occupants' intrinsic thermal preferences and adaptive behaviours in response to an uncomfortable indoor environment, future work could incorporate direct occupant feedback, such as manual thermostat adjustments or override actions, as an additional control input. Integrating this behavioural feedback with CLO and MET may enable the system to infer whether occupants' clothing choices primarily reflect personal preference or respond to persistently low indoor temperatures.

Additionally, while this study focused primarily on temperature as the control variable, future research should extend the framework to multi-objective control strategies that jointly optimize thermal comfort and carbon performance, leveraging the variable-specific carbon sensitivity data established in this study. Furthermore, this study does not explicitly model other dimensions of occupant variation beyond MET and CLO, such as occupancy schedules, arrival and departure patterns, and adaptive behavioural responses. Although the naturally varying occupancy patterns observed in the living lab partially reflect the stochastic characteristics of real-world operation, future research should incorporate a broader range of behavioural factors to further improve the ecological validity and practical applicability of occupant-centric control strategies. While this study demonstrates the feasibility of occupant-centric control using low-cost, general-purpose edge hardware and standard HVAC setpoint interfaces, the scalability of the proposed framework has not yet been quantitatively validated. Practical implementation challenges, including camera coverage requirements for reducing blind spots in complex indoor environments and privacy concerns associated with continuous occupant imaging, require further investigation. Therefore, future studies should evaluate the scalability of the framework across a broader range of building sizes, configurations, and operational contexts. Despite these limitations, integrating dynamic occupant information into the control loop is a cost-effective, scalable strategy that can enhance operational performance without fundamentally altering existing system architectures. Furthermore, the findings provide empirical support for expanding building energy policies beyond equipment efficiency upgrades toward operational, occupant-centric control strategies.

Conclusion

This study proposes a building control framework that makes decisions based on dynamic occupant states and quantitatively evaluates the impacts of these decisions on thermal comfort and carbon emissions during building operation. By estimating thermal comfort using real-time occupant metabolic rate and clothing conditions and integrating these variables into the control logic, the study demonstrates through both simulation and field experiments that the proposed approach produces operational behaviours and environmental outcomes distinct from conventional temperature-based control.

The results indicate that variations in occupant activity exert a greater influence on carbon emissions than changes in CLO, with this effect more pronounced under heating than under cooling conditions. Under identical scenarios, carbon emissions differed by up to 7-fold during heating periods and by approximately 2-fold during cooling periods. Notably, this directional pattern was consistently observed in the test-bed experiments, which most directly parallel the simulation's systematic variation of both MET and CLO. The living lab further demonstrated that incorporating real-time occupant information into control decisions can provide practical benefits under realistic, multi-occupant office conditions. In the living lab, the occupant-centric control strategy reduced carbon emissions by approximately 63% on average across the four-day observation period, compared to conventional temperature-based control. To isolate the effect of the control strategy itself, outdoor weather conditions were matched between the two control modes; under this weather-matched comparison, the reduction remained substantial at approximately 42.8%. This confirms that the observed benefit is not primarily an artifact of weather variability.

Overall, the findings confirm that dynamic human data should not be regarded as a secondary adjustment factor in building control, but rather as a critical input that can substantially influence control decisions and associated energy and carbon outcomes. The results suggest that occupant-centric control can serve as a practical operational strategy that simultaneously maintains thermal comfort and reduces operational carbon emissions. Furthermore, integrating occupant state information into the control loop offers a scalable approach to enhance building sustainability without fundamentally altering existing system architectures.

From a practical standpoint, the proposed occupant-centric control framework can be implemented within existing HVAC infrastructure by modifying the control logic, without requiring substantial hardware retrofits. This offers building operators and facility managers a cost-effective strategy for reducing operational carbon emissions while maintaining thermal comfort. The ability to dynamically adjust system operation in response to occupant states enables more flexible and responsive control, particularly in multi-occupant environments, where static assumptions often lead to over- or under-conditioning.

The findings of this study further suggest that building decarbonization strategies should move beyond conventional system-centric approaches toward occupant-centric operational control. This direction is consistent with emerging user-centered smart building approaches that integrate occupant information to improve thermal comfort, operational efficiency, and sustainable building performance.[41] Integrating occupant-centric control mechanisms into building energy policies and operational guidelines may unlock additional carbon-reduction potential that conventional system-centric frameworks have left unaddressed. Furthermore, future research should extend this framework to multivariable HVAC control systems that incorporate additional parameters, such as airflow, humidity, and inter-system interactions.

Author contributions

Eun Ji Choi: Conceptualization, Writing – original draft, Writing – review & editing. Young Jae Choi: Resources, Data curation. Min Chae Seo: Investigation. Geun Young Yun: Formal analysis. Jin Woo Moon: Project administration, Funding acquisition.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethics approval statements

This study received ethical approval from the Chung-Ang University IRB (Approval No. 1041078-20241114-HR-326) on January 14, 2025. Respondents gave written consent for review and signature before starting experiments.

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Appendix 1. Information on the metabolic rate and the clothing insulation

The information on the metabolic rate and the CLO used in this study is based on the ASHRAE Fundamentals and ISO 9920 standards. The metabolic rate refers to the heat generated by the human body, assuming a body surface area of 1.8 m^2 for an adult male, with 1 met equating to 58.1 W/m^2. Common indoor activities are listed in Table 3. Activity levels are expressed in met, W/m^2, and W/person units, with W/person being the unit used in EnergyPlus. The indoor activities presented in Table 3 range from a minimum of 0.7 met to a maximum of 4.4 met. For the simulations and experiments conducted in this study, activity levels corresponding to the indoor activities in Table 3 were used.

CLO refers to the thermal resistance of all clothing worn by an individual, with 1.0 clo equating to 0.155 m^2K/W. Therefore, CLO values can vary significantly depending on the combination of garments worn. ISO 9920 provides typical CLO values for various clothing combinations, as shown in Table 4. Table 4 illustrates typical indoor clothing combinations with insulation values ranging from 0.3 to 1.0 clo.

Table 3. Indoor activities and metabolic rate.[17]

Activity	Met	W/m^2	W/person
Sleeping	0.7	40	72
Reclining	0.8	45	81
Seated, quiet	1.0	60	108
Standing, relaxed	1.2	70	126
3.2 km/h (0.9 m/s)	2.0	115	207
4.3 km/h (1.2 m/s)	2.6	150	270
Reading, seated	1	55	99
Writing	1	60	108
Typing	1.1	65	117
Filing, seated	1.2	70	126
Filing, standing	1.4	80	144
Walking about	1.7	100	180
Lifting/packing	2.1	120	216
Cooking	1.6 to 2	95 to 115	171 to 207
Housecleaning	2.0 to 3.4	115 to 200	207 to 360
Seated, heavy limb movement	2.2	130	234
sawing (table saw)	1.8	105	189
light (electrical industry)	2 to 2.4	115 to 140	207 to 252
heavy	4.0	235	423
Dancing, social	2.4 to 4.4	140 to 255	252 to 459
Calisthenics/exercise	3 to 4	175 to 235	315 to 423

Appendix 2. Estimation of operational carbon emissions

Operational carbon emissions in this study were quantified based on the energy consumption of building systems during the operational phase. Emissions from multiple greenhouse gases were converted into CO_2-eq using their respective GWP values. The greenhouse gases considered were CO_2, CH_4, and N_2O, in accordance with international reporting standards.

Total operational carbon emissions were calculated by applying energy-source-specific emission factors (f_e) to the consumed energy, as expressed in Equation (3). The energy sources considered in this study were electricity and natural gas (LNG). Carbon emissions associated with fuel consumption were computed using Equation (4), which incorporates fuel consumption, net calorific value, emission factors, and GWP values for each greenhouse gas. Carbon emissions from electricity consumption were calculated using Equation (5), in which electricity consumption was multiplied by the corresponding emission factors and GWP values. Carbon emissions from electricity consumption were calculated following the location-based method of the GHG Protocol Scope 2 Guidance. Since the consumption-based grid emission factor used in this study already reflects the national generation mix and transmission/distribution losses, no additional primary energy factor was applied.

$$CO_2eq = Use\ Energy \times f_e \tag{3}$$

$$CO_2eq = \sum (Fuel_{used} \times Net\ Calorific\ Value \times f_e \times GWP) \tag{4}$$

$$CO_2eq = \sum (Power_{used} \times f_e \times GWP) \tag{5}$$

Where $Fuel_{used}$ denotes the amount of fuel consumption (in Nm^3 or kg), $Power_{used}$ represents the electricity consumption (MWh), f_e is the emission factor (in kgGHG/TJ or kgGHG/MWh).

Emission factors (f_e) for natural gas combustion were adopted from the IPCC 2006 Guidelines for National Greenhouse Gas Inventories.[43] To ensure regional accuracy, electricity emission factors were obtained from the Greenhouse Gas Inventory and Research Center of Korea (GIR).[44] The calculation applied the 2023 consumption-based national grid emission factors, which inherently account for transmission and distribution losses. The GWP values for CO_2, CH_4, and N_2O were set to 1, 28, and 273, respectively, following the IPCC Sixth Assessment Report (AR6).[45] A summary of the emission factors and parameter values used in this study is provided in Table 5.

Table 4. Typical clothing ensembles examples.[42]

Clothing ensembles	Clo
Panties, T-shirt, shorts, light socks, sandals	0.3
Panties, petticoat, stockings, light dress with sleeves, sandals	0.45
Underpants, shirt with short sleeves, light trousers, light socks, shoes	0.5
Underpants, shirt, lightweight trousers, socks, shoes	0.6
Panties, petticoat, stockings, dress, shoes	0.7
Panties, petticoat, shirt, skirt, thick knee socks, shoes	0.8
Panties, shirt, skirt, round neck sweater, thick knee socks, shoes	0.9
Panties, shirt, trousers, jacket, socks, shoes	1.0

Table 5. Emission factors by energy source.

Energy source	GHG	Emission factor (f_e)	Unit
Natural Gas	CO_2	56,100	$kgCO_2/TJ$
	CH_4	5	$kgCH_4/TJ$
	N_2O	0.1	kgN_2O/TJ
Electricity	CO_2	414.9	$kgCO_2/MWh$
	CH_4	0.075	$kgCH_4/MWh$
	N_2O	0.0081	kgN_2O/MWh

Table 6. Average PMV for heating season.

	CLO									
MET	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.7	−0.24	−0.22	−0.20	−0.18	−0.16	−0.15	−0.13	−0.12	−0.11	−0.10
0.8	−0.20	−0.17	−0.15	−0.13	−0.11	−0.10	−0.08	−0.07	−0.05	−0.03
0.9	−0.13	−0.10	−0.08	−0.06	−0.05	−0.02	0.00	0.02	0.03	0.05
1.0	−0.07	−0.05	−0.03	0.00	0.02	0.04	0.06	0.08	0.10	0.11
1.1	−0.05	−0.03	0.00	0.02	0.04	0.06	0.08	0.10	0.11	0.13
1.2	−0.03	−0.01	0.02	0.04	0.06	0.08	0.10	0.12	0.13	0.14
1.3	−0.01	0.01	0.04	0.06	0.08	0.11	0.12	0.13	0.14	0.15
1.4	0.01	0.03	0.05	0.08	0.10	0.12	0.13	0.14	0.15	0.15
1.5	0.03	0.06	0.08	0.11	0.12	0.13	0.14	0.15	0.16	0.16
1.6	0.05	0.08	0.10	0.12	0.14	0.15	0.15	0.16	0.16	0.17
1.7	0.08	0.11	0.13	0.14	0.15	0.16	0.17	0.17	0.17	0.18

Appendix 3. Simulation-based PMV distribution across occupant states

The mean PMV values across the simulated MET–CLO combinations are presented in Tables 6 and 7 for the heating and cooling seasons, respectively. As described in the main text, the heating period maintained a mean PMV of 0.04 with a standard deviation of 0.11, showing a slightly wider distribution than the cooling period. In contrast, the cooling period exhibited a mean PMV of 0.01 and a standard deviation of 0.01, remaining consistently within the acceptable comfort range. This stable performance reflects the controlled simulation environment, where external disturbances were minimized, enabling consistent thermal comfort control across different occupant states.

Table 7. Average PMV for cooling season.

	CLO									
MET	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.7	−0.02	−0.02	−0.01	−0.01	−0.01	−0.01	0.00	0.00	0.00	0.00
0.8	−0.01	−0.01	−0.01	−0.01	0.00	0.00	0.00	0.00	0.01	0.01
0.9	−0.01	−0.01	0.00	0.00	0.01	0.01	0.01	0.01	0.02	0.02
1.0	0.00	0.00	0.01	0.01	0.01	0.02	0.02	0.02	0.03	0.03
1.1	0.00	0.00	0.01	0.01	0.01	0.02	0.02	0.03	0.03	0.02
1.2	0.00	0.01	0.01	0.01	0.02	0.02	0.03	0.03	0.02	0.02
1.3	0.01	0.01	0.01	0.02	0.02	0.03	0.03	0.02	0.02	0.02
1.4	0.01	0.01	0.02	0.02	0.02	0.03	0.02	0.02	0.02	0.01
1.5	0.01	0.02	0.02	0.03	0.03	0.02	0.02	0.02	0.01	0.01
1.6	0.01	0.02	0.03	0.03	0.02	0.02	0.02	0.01	0.01	0.01
1.7	0.02	0.02	0.02	0.02	0.02	0.02	0.01	0.01	0.01	0.01